Experiments

Stat 256, Causality

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Identification vs. Estimation

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For each identification strategy we discuss:

1. the identification assumption
2. information from outside the data supporting it
3. testable implications (limited)
4. estimators
5. sensitivity to violations

Classical Randomized Experiment

Setup:

- Units: $i = 1, \ldots, N$
- Treatment: $D_i \in \{0, 1\}$, randomly assigned
- Potential outcomes: Y_{0i}, Y_{1i}
- Observed outcome: $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$ (consistency)
- $N_1 = \sum_{i=1}^{N} D_i, \quad N_0 = N - N_1$

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Random assignment can take several forms:

- Complete randomization: Exactly N_1 units treated (draw without replacement)
- Simple (Bernoulli) randomization: Each unit independently assigned with $\Pr(D_i = 1) = p$

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This is the identification assumption for experiments. Let's see what it buys us.

Identification of ATE Under Experiments

Quantity of Interest:

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\mathbb{E}[Y_{1i}] - \mathbb{E}[Y_{0i}] &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \quad \text{(by ignorability)} \\
&= \mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] \quad \text{(by consistency)} \\
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Practice these two steps. They are the core of every ID argument we will see.

Identification of Full Distributions

With $(Y_1, Y_0) \perp\!\!\!\perp D$, we identify more than means.

Identification Result

Let $F_{Y_d}(y)$ denote the CDF of Y_d. Then

$$\begin{aligned} F_{Y_0}(y) &= \Pr(Y_0 \leq y) = \Pr(Y_0 \leq y \mid D = 0) = \Pr(Y \leq y \mid D = 0) \\ F_{Y_1}(y) &= \Pr(Y \leq y \mid D = 1) \end{aligned}$$

So the effect at any quantile $\theta \in [0, 1]$ is identified:

$$\alpha_\theta = Q_\theta(Y_1) - Q_\theta(Y_0) = Q_\theta(Y \mid D{=}1) - Q_\theta(Y \mid D{=}0)$$

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Note: the *quantile of the effect* $Q_\theta(Y_1 - Y_0)$ is **not** identified without further assumptions — the difference of quantiles is not the quantile of the difference.

SATE and PATE

- **SATE**: $\tau_{SATE} = \frac{1}{N}\sum_{i=1}^{N}(Y_{1i} - Y_{0i})$
  - Inferences limited to units in the study
  - Resampling notion can vary:
    - *Design-based*: $\boldsymbol{X}$, POs fixed; only treatment assignment is re-randomized
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- **PATE**: $\tau_{PATE} = \mathbb{E}[Y_{1i} - Y_{0i}]$ over a superpopulation
  - Generalizes the above to also redraw units from a larger superpopulation/DGP
- If sampling is random from the population, $\mathbb{E}[\text{SATE}] = \text{PATE}$. We are often uninterested in distinguishing them.

Variance of the DIM: PATE

Start from the variance of the mean difference:

$$\mathbb{V}(\hat{\tau}) \;=\; \mathbb{V}(\bar{Y}_1) + \mathbb{V}(\bar{Y}_0) - 2\mathrm{Cov}(\bar{Y}_1, \bar{Y}_0)$$

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For the PATE (superpopulation view): treated and control units are independent draws, so $\mathrm{Cov}(\bar{Y}_1, \bar{Y}_0) = 0$:

$$\mathbb{V}_{\text{Neyman}}(\hat{\tau}) = \frac{\sigma^2_{Y_1}}{N_1} + \frac{\sigma^2_{Y_0}}{N_0}$$

This is the **Neyman variance**.

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Plug-in estimator: replace population variances with sample variances:

$$\widehat{SE}(\hat{\tau}) = \sqrt{\frac{S_1^2}{N_1} + \frac{S_0^2}{N_0}}, \qquad S_d^2 = \frac{1}{N_d - 1} \sum_{i:D_i=d} (Y_i - \bar{Y}_d)^2$$

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- $\mathbb{V}(\bar{Y}_d)$ is *smaller* than $\sigma^2_{Y_d}/N_d$ by a finite population correction factor $(1 - N_d/N)$
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Working through the finite-population sampling algebra, these combine to give:

$$\mathbb{V}_{SATE}(\hat{\tau}) = \frac{\sigma^2_{Y_1}}{N_1} + \frac{\sigma^2_{Y_0}}{N_0} - \frac{\sigma^2_{\tau}}{N}$$

where $\sigma^2_{\tau} = \mathrm{Var}(\tau_i) \geq 0$.

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So the Neyman variance is **conservative** for the SATE. The gap is σ^2_{τ}/N: zero if effects are constant, and shrinking with N regardless.

Regression to Estimate ATE

Estimator (Regression with Experiments)

$$
\begin{aligned}
Y_i &= D_i Y_{1i} + (1 - D_i) Y_{0i} \\
&= Y_{0i} + (Y_{1i} - Y_{0i}) D_i \\
&= \underbrace{\bar{Y}_0}_{\alpha} + \underbrace{(\bar{Y}_1 - \bar{Y}_0)}_{\hat{\tau}} D_i + \underbrace{\{(Y_{0i} - \bar{Y}_0) + D_i[(Y_{1i} - \bar{Y}_1) - (Y_{0i} - \bar{Y}_0)]\}}_{\varepsilon_i}
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- The residual ε_i has different variances under $D = 1$ and $D = 0 \Rightarrow$ heteroskedasticity.
- We need a heteroskedasticity-robust SE.

The Sandwich Variance Estimator

For OLS with $\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$:

$$\begin{aligned}\widehat{\mathbb{V}}(\hat{\beta} \mid \boldsymbol{X}) &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \, \widehat{\mathrm{Var}(Y \mid \boldsymbol{X})} \, \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} \\ &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\hat{\Omega}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\end{aligned}$$

where $\hat{\Omega} = \mathrm{diag}(\hat{\omega}_1, \ldots, \hat{\omega}_N)$ estimates the conditional variance of each Y_i.

The choice of $\hat{\omega}_i$ defines the variant:

HC0: $\hat{\omega}_i = \hat{\varepsilon}_i^2$

HC1: $\hat{\omega}_i = \frac{N}{N-p}\hat{\varepsilon}_i^2$ (global degrees-of-freedom correction)

HC2: $\hat{\omega}_i = \frac{\hat{\varepsilon}_i^2}{1-h_{ii}}$ (leverage correction, h_{ii} = ith diagonal of $\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$)

With HC0, the "meat" of the sandwich becomes:

$$\boldsymbol{X}'\hat{\Omega}\boldsymbol{X} = \sum_{i=1}^{N} \hat{\varepsilon}_i^2 \, \boldsymbol{x}_i\boldsymbol{x}_i'$$

where $\hat{\varepsilon}_i = Y_i - \boldsymbol{x}_i'\hat{\beta}$ and $\boldsymbol{x}_i$ is the ith row of $\boldsymbol{X}$.

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Sandwich for the Experimental Regression

The $[2,2]$ element of $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\text{diag}(\hat{\varepsilon}_i^2)\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}$ is just

$$\widehat{\mathbb{V}}(\hat{\tau}) = (\tilde{D}'\tilde{D})^{-1}\tilde{D}'\text{diag}(\hat{\varepsilon}_i^2)\tilde{D}(\tilde{D}'\tilde{D})^{-1}$$

where $\tilde{D}$ is the vector $D_i - \bar{D}$. This sorts and splits into two pieces:

$$= \frac{\sum_{i:\, D_i=1} \hat{\varepsilon}_i^2\, \tilde{D}_i^2}{\left(\tilde{D}'\tilde{D}\right)^2} + \frac{\sum_{i:\, D_i=0} \hat{\varepsilon}_i^2\, \tilde{D}_i^2}{\left(\tilde{D}'\tilde{D}\right)^2}$$

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Noting that $\tilde{D}_i = N_0/N$ for treated, and $-N_1/N$ for controls, and that $\tilde{D}'\tilde{D} = N_1 N_0/N$, after some tedious algebra we get this to equal

$$= \frac{\widehat{\text{Var}}(Y|D=1)}{N_1} + \frac{\widehat{\text{Var}}(Y|D=0)}{N_0}$$

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$$= \frac{\widehat{\text{Var}}(Y|D=1)}{N_1} + \frac{\widehat{\text{Var}}(Y|D=0)}{N_0}$$

Further, if you construct $\widehat{\text{Var}}(Y|D=d)$ with the $N_d - 1$ correction then you have the HC2 correction and this matches Neyman exactly.

Example: Effect of Job Training on Earnings

- Treatment: $N_1 = 7{,}487, \quad \bar{Y}_1 = \$16{,}199, \quad S_1 = \$17{,}038$
- Control: $N_0 = 3{,}717, \quad \bar{Y}_0 = \$15{,}040, \quad S_0 = \$16{,}180$

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All exactly the same thing:

- DIM with Neyman SE
- Welch (unequal variance) t-test
- `lm(earnings ~ D)` with HC2 SEs via `sandwich::vcovHC(fit, type="HC2")`

Fisher's Exact Test

You already know the large-N test:

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Fisher's Exact Test uses a stronger hypothesis:

$$H_0 : Y_{1i} = Y_{0i} \ \ \forall\, i \qquad \text{(sharp null of no effect)}$$

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$$H_0 : \mathbb{E}[Y_1] = \mathbb{E}[Y_0], \quad H_1 : \mathbb{E}[Y_1] \neq \mathbb{E}[Y_0] \qquad \text{(weak null)}$$

Fisher's Exact Test uses a stronger hypothesis:

$$H_0 : Y_{1i} = Y_{0i} \ \ \forall\, i \qquad \text{(sharp null of no effect)}$$

Key idea: Under the sharp null, we "observe" all potential outcomes — so we can compute the test statistic under every possible treatment assignment.

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Procedure:

1. Compute $\hat{\tau}_{\text{obs}}$ from actual assignment
2. For every (or many) alternative assignments $\omega \in \Omega$, compute $\hat{\tau}(\omega)$ using the filled-in PO schedule
3. p-value $= \Pr_\omega\big(|\hat{\tau}(\omega)| \geq |\hat{\tau}_{\text{obs}}|\big)$

Fisher's Exact Test: Small-Sample Example

i	Y_{1i}	Y_{0i}	D_i
1	3	?	1
2	1	?	1
3	?	0	0
4	?	1	0
$\hat{\tau}_{\text{obs}}$			1.5

What do we know given $H_0 : Y_{1i} = Y_{0i} \ \forall i$?

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Sharp null fills in the full schedule. Now enumerate $\binom{4}{2} = 6$ assignments.

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i	Y_{1i}	Y_{0i}	D_i	ω_1	ω_2	ω_3	ω_4	ω_5
1	3	3	1	1	1	0	0	0
2	1	1	1	0	0	1	1	0
3	0	0	0	1	0	1	0	1
4	1	1	0	0	1	0	1	1
$\hat{\tau}(\omega)$			1.5	0.5	1.5	-1.5	-0.5	-1.5

$\Pr\big(|\hat{\tau}(\omega)| \geq 1.5\big) = 4/6 \approx 0.67.$

Design-based: fixed POs, D varies under re-randomization. No distributional assumptions.

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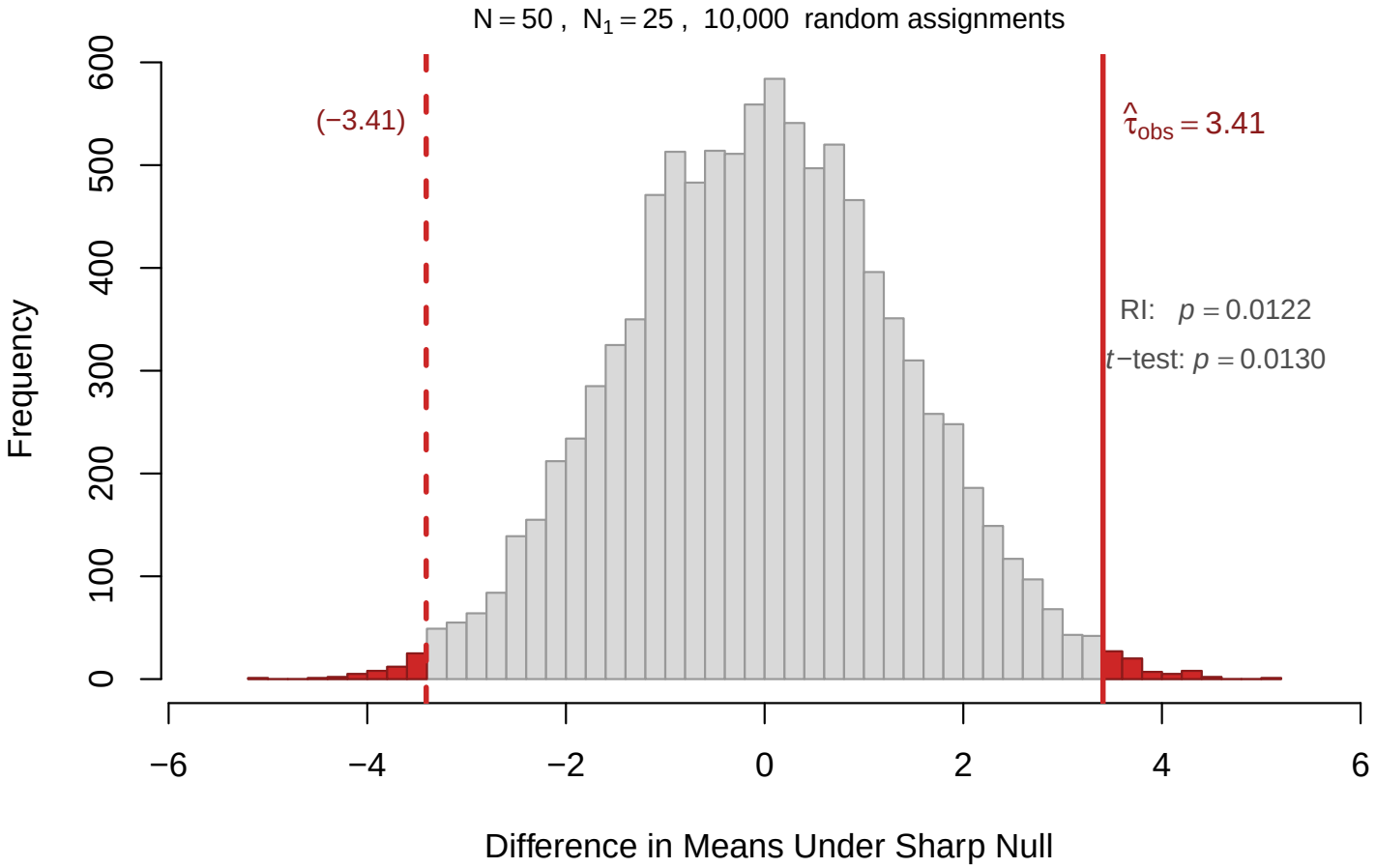
Solution: sample from Ω.

- Re-draw treatment assignment many times (e.g. 10,000)
- Compute $\hat{\tau}(\omega)$ each time under the sharp null
- Compare observed $\hat{\tau}$ to this simulated null distribution

Randomization Inference in Practice



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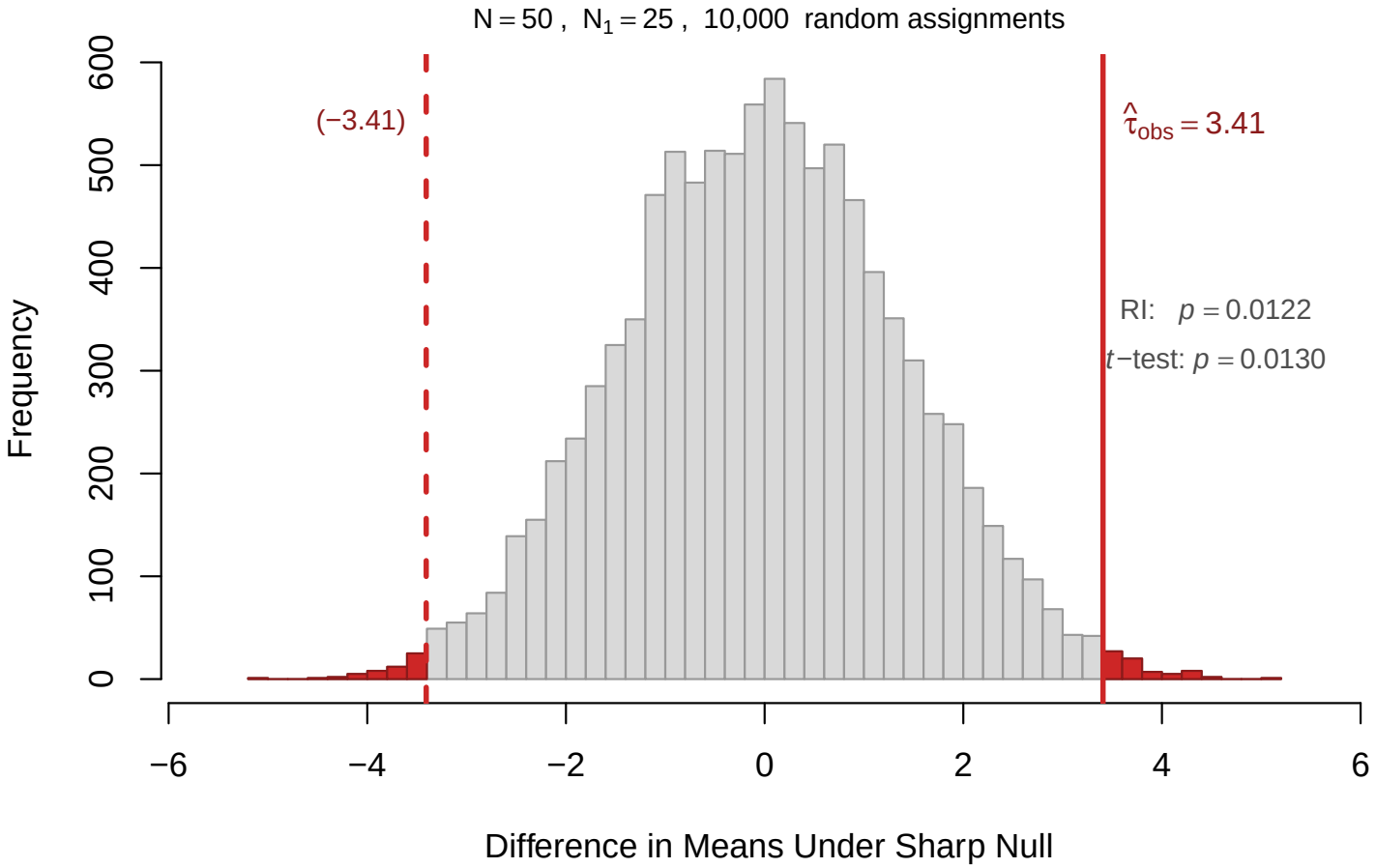

Histogram: 10,000 re-randomizations under the sharp null.
Solid line: observed $\hat{\tau}$. Shaded: values at least as extreme $\Rightarrow$ p-value.

How Do These Inferences Differ?

Finite-sample exactness.

- The RI p-value is exact under the sharp null — no CLT required.
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- The sharp null says $Y_{1i} = Y_{0i}$ for *every* unit.
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RI vs. the bootstrap.

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Bottom line: RI most valuable in small samples where CLT inference is suspect.

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Everything so far assumed (conditionally) independent observations. Often violated:

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The dangerous case is **both at once**: a cluster-level component in Y *and* no within-cluster treatment variation to cancel it out.

Cluster-Robust SEs

The cluster-robust ("CR") sandwich estimator:

$$\widehat{\mathbb{V}}_{CR}(\hat{\beta}) = (\boldsymbol{X}'\boldsymbol{X})^{-1} \left(\sum_{c=1}^{C} \boldsymbol{X}_c' \hat{\varepsilon}_c \, \hat{\varepsilon}_c' \boldsymbol{X}_c \right) (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

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- With few clusters ($C \lesssim$ 30–50), CR-SEs are downward biased. Alternatives: wild cluster bootstrap (`fwildclusterboot` in R), randomization inference at the cluster level, or aggregate to cluster means.
- If there is *no* cluster-level component in Y, or if treatment varies within cluster so the component cancels, then HC0 is fine — CR-SEs buy nothing and widen CIs.

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So the goal is to **reduce variance**. Two levers:

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2. **Analysis**: covariate adjustment (remove imbalance after the fact)

Both exploit the same idea: a pre-treatment covariate X that predicts Y can absorb residual variation.

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Rule of thumb: block on the strongest predictor of Y you have. A pre-treatment version of the outcome is ideal. Blocking on something unrelated to Y costs nothing but gains nothing.

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Best practice: "Block on what you can, adjust for what you can't." The two

Takeaways

- **Randomization = ignorability.** Identification uses two steps: ignorability, then consistency.
- **DIM = Neyman = Welch t = OLS with HC2.**
- **Randomization inference** is exact in finite samples and design-based. Rejecting the sharp null via DIM is close to rejecting the weak null.
- **Clustering** inflates variance and/or breaks SE estimation. Cluster-robust SEs handle it with enough clusters.
- **Blocking and covariate adjustment** reduce variance — unbiasedness alone is not the goal. Block on what you can, adjust for what you can't.
- **Next**: What if we can't randomize?

Appendix

Why Neyman Is Conservative for SATE

Recall the exact variance:

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Either way, CIs based on Neyman SEs have *at least* nominal coverage for the SATE.